

ADITYA SHIRSATRAO

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PROFESSIONAL SUMMARY

SDE-oriented Full-Stack & Systems Engineer building low-latency backends, distributed streaming platforms, and production AI systems. Authored C++17 order matching engine (10.9M orders/sec, price-time priority) and Kafka+Spark pipeline (10K events/sec, p99 <500ms). Shipped 3 hackathon-winning products (600--800+ teams) and merged 50+ PRs at Google, Microsoft, Apache, Hedera. 366 LeetCode (104E/176M/86H) + 50 SQL. GDG AI/ML Lead mentoring 200+ students.

EDUCATION

N. K. Orchid College of Engineering and Technology Solapur, MH
B.Tech -- Artificial Intelligence and Data Science — CGPA: 8.5/10 — Expected 2027
Class XII (HSC): 90% -- Sarosh Junior College, Aurangabad | **Class X (ICSE):** 90% -- St. Thomas EM School, Solapur Aug 2023 -- Present

EXPERIENCE

Fullstack Web Development Intern Jul 2025 -- Jan 2026
Unified Mentor Pvt. Ltd. -- Production Systems & Distributed Platforms Remote

- Built 6 production-grade platforms: **AI code review** (Kafka+Redis, -45% latency via parallel workers & AST vector search), **real-time collaborative IDE** (WebSockets/Socket.IO/Redis pub/sub, CRDT, Monaco Editor, sub-100ms sync), **distributed ML inference** (+60% throughput via batching/caching, Prometheus), **event streaming analytics** (Kafka+Spark), **AI interview platform** (Whisper+LLM), and **microservices e-commerce** (Docker/Kubernetes, payment & inventory consistency).
- Stack: React, Next.js, Node.js, FastAPI, PostgreSQL, Redis, Kafka, Docker, Kubernetes, WebSockets, OAuth, GitHub Actions CI/CD. Shipped with observability and multi-env deployments.

AI/ML Lead Sep 2025 -- Present
Google Developer Groups on Campus (GDG-NKOCET) Solapur

- Lead AI/ML for 200+ students; workshops on system design, distributed systems & production ML. Mentored juniors on Git/GitHub, backend architecture, and shipping. Class Representative (2023--27, 120+ cohort).

TECHNICAL SKILLS

Languages: Python, C++17, Java, JavaScript/TypeScript, Go, SQL | **Backend:** FastAPI, Node.js, PostgreSQL, Redis, Celery, Docker, Kubernetes, Kafka, Spark, Microservices, REST, JWT/RBAC, CI/CD
Frontend: React.js, Next.js, Tailwind, WebSockets/Socket.IO | **ML/Systems:** PyTorch, HuggingFace, RAG/FAISS, Whisper, BERT 92%+ | **Core:** DSA 366 (86H) + 50 SQL, System Design, Distributed Systems

PROJECTS

Order Matching Engine -- Low-Latency C++17 Limit Order Book | GitHub | C++17, FastAPI, WebSocket Aug 2025

- Deterministic matching core with **price-time priority & FIFO** within level; supports LIMIT/MARKET/STOP with GTC/IOC/FOK, partial fills, cancellations. Zero external deps, trade tape + depth callbacks; **10.9M+ matches/sec** on bench harness (µs/order). FastAPI+WebSocket bridge: `POST /orders`, `GET /depth`, `WS /ws` live ticks. Dockerized with CI.

SentinelX -- AI Cybersecurity Platform | GitHub | FastAPI, DistilBERT, Whisper, Celery, Redis, PostgreSQL May 2026

- Multi-channel threat detection (email/SMS/voice) with DistilBERT NLP + Whisper STT + Qwen 2.5-7B enrichment; **4-component weighted risk engine** (NLP-0.35 + Behavior-0.25 + URL-0.20 + Reputation-0.20). Async Celery+Redis pipeline, JWT/RBAC, Alembic migrations. **1st Place, Hack to Future 3.0 (600+ teams, 50k), shipped in 36h.**

Real-Time Clickstream Pipeline -- Kafka + Spark Structured Streaming | GitHub | Kafka, Spark, Docker Aug 2025

- Producer → Kafka topic `clicks` → Spark Structured Streaming: windowed sessionization, top-page ranking, spike anomaly detection with watermarking for late data. Sustained **10K events/sec, p99 <500ms**; `docker compose` up for Kafka/Zookeeper/Spark. Benchmark mode for throughput/latency.

OPEN SOURCE & ADDITIONAL PROJECTS

50+ PRs merged -- Google OSS-Fuzz, Microsoft VibeVoice, Apache DataFusion, Hedera Hiero SDK (10k+ devs impacted). Hiero PR #1030: -30% boilerplate; Super Impress PR #77: +15% CLS; DataFusion v43 JOIN optimizer repro.
Tracelify (PyPI: tracelify-sdk) -- 6-lang SDK + FastAPI/Redis/RBAC + Bedrock LLM reports; 1st Runner-Up Orchathon 2K26 (800 teams). **BERT Multilingual** (92%+, 104 langs, HuggingFace) -- IEEE ICCTWC 2026 first-author. **Nyaya AI** (EasyOCR FIR, 500+ IPC) -- National Runner-Up (800 teams).

ACHIEVEMENTS & CERTIFICATIONS

3× National Hackathon Winner: 1st Place Hack to Future 3.0 (600 teams), 1st Runner-Up Orchathon 2K26 (800 teams), 1st Runner-Up National Legal-Tech Hackathon (800 teams).
366 LeetCode + 50 SQL (86 Hard) — **NPTEL Programming in Java Elite+Gold (91/100)** — Class Rep 120+ — 40+ repos, 105 merged PRs, 32 orgs.
Certifications: Kaggle AI Agents Intensive (2025), JPMorgan Chase SE Virtual Experience, Electronic Arts SE Virtual Experience, AWS Bedrock, Joy of Computing (NPTEL).